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REGULAR PAPER

Persian Aut emotional speech database: machine learning-based evaluation and cost-efficient labeling approach

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Abstract
Speech emotion recognition (SER) plays a pivotal role in different areas, such as the evolution of automated customer service systems, enabling a more sophisticated understanding of user sentiments. In the context of the Persian language as a low-resource language, the absence of natural emotional speech datasets poses a challenge for developing effective systems. This paper addresses this gap by engaging 28 non-professional university students in creating a semi-natural dataset as a basic representative of real-world interactions. This dataset called AutESD having four emotions of neutrality, happiness, anger, and sadness could also be adapted to be used for on-line academic evaluations in future since we have gathered it from both male and female students in the age range of 23 to 30. The main advantages of AutESD are its semi-natural characteristic, larger number of speakers compared to previous Persian datasets, and its low cost in the collection procedure. Moreover, we propose and develop a labeling software to streamline the labeling process which uses 16 persons including both speech experts and non-experts to label the emotion classes remotely. The inter-rater reliability analysis yielded Fleiss' Kappa values between 0.70 and 1.00, indicating substantial to almost perfect agreement among annotators. It establishes a framework to optimize both labeling expenses and time as well as accuracy. Subsequently, to show the efficiency of AutESD in SER task, we evaluate it utilizing four machine learning models of K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Deep Neural Network (DNN)-based model which uses a Convolutional Neural Network (CNN) architecture, and a transformer. The results show that AutESD achieved Unweighted-Average-Recall (UAR) of up to 79% using CNN by MFCC features on the test dataset using 5-fold cross validation in a speaker independent scenario.

Keywords Speech emotion recognition (SER) · Persian language · Low-resource dataset · Machine learning · Semi-natural dataset

1 Introduction

Understanding emotions conveyed through speech signals is essential for enhancing Human-Computer Interaction (HCI), as speech serves as a powerful medium for expressing emotions and attitudes, underscoring the importance of accurately identifying emotional content in speech [1–3]. In recent years, with the increasing demand for artificial intelligence, especially due to the fourth industrial revolution, the importance of the number of data mainly for low-resource languages [4], and their quality to train neural networks has been a challenge [5]. Researchers endeavor to address this issue by increasing model complexity, employing advanced techniques such as deep Convolutional Neural Networks (CNN-Based) and Long Short-Term Memory (LSTM-Based) neural networks [6–9], using Transformers [10], or joint optimized approaches based on appropriate deep networks such as autoencoders [11]. On the other hand, due to the fact that speech is the easiest way for humans to communicate, the demand for voice datasets for intelligent and convenient communication is increasing day by day [12]. Speech carries a lot of data in which

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emotion is one of the most important ones. Extraction of emotions from speech can enhance the performance of the neural network models in the speech processing field [13]. Although datasets are evolving, the disparity between datasets and real-world scenarios remains apparent, particularly in low-resource languages [14, 15]. Datasets such as Shemo [16] and [17] have been developed for speech in the Persian language. However, Shemo is a semi-natural dataset that is created by collecting speech from radio programs and it is far away from the real-world scenarios. Also, PersianESD is made up of simulated data in which only two professional actors are used. Not only does it suffer from the limited number of speakers, but it also suffers from a considerable difference with natural data.

Phone call quality monitoring, checking the emotion of the caller and forming a queue based on the caller's emotion in a call center in order to form a more effective queue especially in urgent conditions are among the most important applications of SER. In the context of online education, these datasets prove invaluable for evaluating the quality of virtual classes, offering real-time insights into students' sentiments which helps institutes to enhance their assessment methodologies. Also, these tasks need natural emotions in at least four categories: neutrality, happiness, sadness, and anger. Previous datasets like Shemo and PersianESD were collected by professional actors, and they are different from natural data.

In this paper, we have created a new Persian dataset for speech emotion recognition called AutESD (AUT emotional speech dataset). To this end, we have used 28 native Persian-speaking students (17 male and 11 female) in order to collect a semi-natural dataset in neutrality, happiness, sadness, and anger categories by the use of 32 Persian sentences, trying to enhance this field of study in the Persian language. Our dataset contains 1428 audio data uttered by 28 Persian speakers in 2 h and 2 min in four different emotions with a sample rate of 44.1 kHz. This is a relatively larger Persian dataset in SER task. Moreover, we propose a new labeling method and software accelerating data collection, reduce labeling costs, and improve the quality of emotion annotations. However, using non-professional speakers has two advantages over other datasets especially the acted or simulated one: (1) It would be costly gathering a large dataset from professionals as done in the simulated class, (2) the semi-natural dataset could be adapted in real-world scenarios with native speakers more easily in comparison with those gathered from actors as professionals. One of the main applications of this new dataset which have been recorded from students' voices as non-professional speakers is making it useful for academic purposes such as automatically detecting the emotions of students while asking questions in on-line classes or webinars which are very popular these days. For instance, we can evaluate the quality of how the main speaker presents the subject by automatically detecting the emotions of participants. In other words, if we detect some kind of anger as the emotion of the participant, it shows that he/she is not satisfied with the quality of the webinar. This evaluation method is much more accurate than traditional evaluation forms because it is a dynamic evaluation method which could determine both the exact time and the subject (event) the students were not happy with the class, or vice versa. Thus, it could be much more helpful for the lecturers to improve their speech. It could also be used in call centers to be prioritized according to the event importance as we mentioned before. Figure 1 shows the overview of the work we have performed for our data collection, the proposed labeling procedure, and evaluation for SER task.

This paper is organized as follows. In Sect. 1.1, we summarize the contributions of our new Persian dataset for SER purposes. In Sect. 2, we present the related works. Section 3 describes the details of our created dataset as well as our proposed labeling method. In Sect. 4, we explain a preview of the data. Section 5 presents the data evaluation for SER by using some machine learning methods. Finally, we conclude the paper in Sect. 6.

1.1 Contributions

The main contributions of our work compared to previous Persian SER datasets such as Shemo and PersianESD are as follows:



Fig. 1 Labeling procedure. Dataset Discovery and Text Collection: This step refers to search, collect and evaluation transcripts. Timing and Equipment Configuration refers to coordinating with students and preparation recordings such as acoustic room, and recording systems configurations. In the Recording step, students are prepared (They write their names on a paper as consent and read the emotional text to get prepared for recording); then, their emotional speeches are recorded. Labeling and Speech Labeling are referred to developing a labeling application (Sect. 3.6.1) and the provided application is given for the final emotion labeling. Also, in the Speech Labeling step, we confirm both the audio qualities and their labels. In the last step (Classifiers Application and Visualization), three classifiers are trained to visualize

- Creating a relatively larger new Persian dataset for speech emotion recognition called AutESD containing four emotions of neutrality, happiness, sadness, and anger: it is made up of 32 Persian sentences recorded from 28 native Persian-speaking students (17 male and 11 female),
- The semi-natural nature of the dataset: it helps adapting it in real-world scenarios with native speakers more easily in comparison with those gathered from actors as professionals such as Shemo and PersianESD. One of the main useful applications of this new dataset is to improve the students' evaluation system in academic environments, i.e. automatically detecting the emotions of students while asking questions in on-line classes or webinars to predict their satisfaction levels,
- Our dataset has been gathered in a balanced manner in all four emotions,
- The low-cost labeling software and improved reliability: we propose a new labeling method and software that accelerates data collection, reduce labeling costs, and improve the quality of emotion annotations. The software features are as follows:
  - Providing a user-friendly interface,
  - Supporting multi-user and remote annotation, enabling the involvement of more annotators,
  - preventing duplicate labeling,
  - Using a two-level emotion tagging system (Strong and Weak) with a "None" option for ambiguous cases. Well-organized outputs in CSV, Excel, and ZIP formats simplify data aggregation and subsequent analysis, ensuring consistent, reliable, and reproducible emotion labeling.

2 Related works

Audio and speech processing applications, especially speech emotion recognition, has become very important and challenging in this era, as an efficient and effective tool in the fourth industrial revolution. As the main examples of the usage of these systems, we can mention the development of intelligent systems [1], human-computer interaction (HCI) and music retrieval [18], brain-computer interface (BCI), robotics [14, 19], safe smart houses, and healthcare applications [1, 20, 21]. Moreover, it is worth stating that real-time applications such as human behavior evaluation, virtual reality, and emergency call centers to analyze the emotional state of speakers [22] are also among important usages. Increasing the quality of user experience in workplaces through interactive robots [19, 23], understanding the emotions and transferring them for people with hearing and speech problems [24], examining mental health [5], and development of intelligent multilingual systems [25] could also be mentioned as real-time applications of SER. In order to guarantee its performance, we require sufficient and reliable data that is comparable to the target environment and have high quality like other fields/applications of artificial intelligence [26, 27]. Recently, numerous data have been published for speech emotion processing. Emotional speech data are classified into three categories: simulated, semi-natural, and natural data [28].

2.1 Acted-simulated data

In this dataset, professional or semi-professional actors play roles to produce emotional speech. These actors can be professional or amateur theater actors. This kind of emotional data are usually expressed with a lot of exaggerations, and are generally different from the natural emotional speech used in everyday talks [27, 29, 30]. Almost 60% of emotional speech data are from this category up to now [31]. In addition to using actors, these data can be collected from conversations between doctor-patient and audio recordings in public centers [30]. PersianESD [17], BanglaSER [32], and EMO-DB [33] are examples of these databases.

2.2 Semi-natural data

In this dataset, speakers are first in the condition of arousing emotions, and then, they start to perform the speech [27, 30]. Actually, these emotional speeches are collected by creating artificial emotional states [31]. Although these data are not completely natural, this type of database is more natural than the actor-based database [27, 30, 31]. IEMOCAP [34], Shemo [16], and SES [35] are some examples of these databases.

2.3 Natural data

This dataset is usually collected from natural conversations such as television debates, live television and radio programs, conferences, and phone calls. Although these data are very similar to real emotions and talks in everyday conversations, collecting them costs a lot. Also, it is impossible to use many of these data due to their special and strict privacy policies [27]. While these data are the most real emotional data, it is difficult to collect this dataset due to the difficulty in recognizing and classifying emotions [29]. Natural emotional speech dataset is usually collected from public conversations, call center conversations [29, 30], recording mother-child and doctor-patient conversations [30], podcasts [26], etc.

2.4 Data collection methods

The quality of the dataset is directly related to the collection method. In order to reduce the impacts of unwanted noises, data collection takes place in an acoustic room [32]. Also, the correct use of audio equipment can improve the quality of data. After audio data collection, the data labels are evaluated again by experts or native speakers. To evaluate audio data labels, a voting form is given to each voter so that they can vote about the emotions in the audio after listening to each file [34]. This work is done by a number of men and women of different ages. The audio file with the majority votes is chosen to get that main label. Table 1 presents the specifications of some of the existing datasets. Aut-ESD refers to the dataset we have created in this research work.

3 Dataset description

3.1 Text

The text of this dataset is collected and arranged according to the Shemo [16] dataset. Tables 2 and 3 show the texts which are uttered by male and female speakers. According to these tables, speakers must utter two sentences (texts with "yes" requirement) from each emotional class, and they can choose at least one sentence from the other two sentences (texts with "no" requirement). Each speaker articulates the selected text at least twice, and the recording is done on different days (sessions) to remove the effects of the speaker's personal feelings. As a result, at least four recordings have been made by each speaker for each text. In addition, we chose the sentences carefully to make an almost natural emotional scenario for the speakers so that they can imitate them with the correct emotion we look for.

Table 1 Comparing the specifications of some existing datasets

Accessibility	# Of audio	Sample rate	Gender size	Type	Emotions	Size	Language	Specification
Free for education	5255 Scripted, 4784 Spontaneous	48 kHz	5 Females, 5 Males	Semi-Natural	Happiness, Anger, Sadness, Frustration, Neutral	12 h	English	IEMOCAP [24]
Free	535	48 kHz	5 Female, 5 Male	Acted	Anger, Boredom, Disgust, Fear, Happiness, Sadness, Neutral	-	German	EMO-DB [22]
Free	1467	44.1 kHz	17 Females, 17 Males	Acted	Angry, Happy, Neutral, Sad, Surprise	1 h 29 min	Bangla	Bangla SER [23]
Free for education	252	44.1 kHz	1 Females, 1 Males	Acted	Anger, Disgust, Fear, Happiness, Sadness, Neutral	-	Persian	PersianESD [5]
Free	3000	44.1 kHz	31 Female, 56 Males	Semi-Natural	Anger, Fear, Happiness, Sadness, Surprise, Neutral	3 h 25 min	Persian	Shemo [16]
-	1200	-	5 Female, 5 Male	Semi-Natural	Neutral, Anger, Sadness, Happiness, Surprise	50 min	Persian	SES [25]
Free	1428	44.1 kHz	11 Females, 17 Males	Semi-Natural	Anger, Sadness, Happiness, Neutral	2 h 2 min	Persian	Aut-ESD (Ours)

Table 2 Sentences for male speakers

N	Sentences (for male speakers)	R	E	ID
1	واقعا که خجالت داره، با این سن و سال هنوز هم دست بردار نیستی؟ vaqeʔæn ke xedʒalæt dare ba ʔin senno sal hænuzæm dæst bærdar nisti	Yes	A	A1
2	اینم بدون که اگه در این مورد کوتاهی کنی ازت شکایت می‌کنم کلانتر. ʔinæm bedun ke ʔæge dær ʔin mored kutahi koni ʔæzæt ʃekayæt mikonæm kælantær	No	A	A3
3	یاشو برو خدا روزیت رو جای دیگه حواله بکنه، یاشو برو جانم، یاشو برو. paʃo boro xoda ruzito dʒaye dige hævale bokone paʃo boro dʒanæm paʃo boro	No	A	A4
4	تو غلط کردی رفتی خونه رو فروختی، من خودم بیشتر از اینا می‌فروختمش. to qælæt kærdi ræfti xunæ ro furuxti mæn xodæm biʃtær ʔæz ʔina mifruxtæmeʃ	Yes	A	A2
5	دکتر بهشتی به حوزه میره و در اونجاست که برای ادامه تحصیل راهی قم می‌شه. doktor beheʃti be hoze mire væ dær ʔundʒast ke bæraye ʔedame tæhsil rahiye qom miʃe	Yes	N	N1
6	در این چند روزی که اینجاییم هیچ اتفاق غیر عادی نیوفتاده. dær ʔin tʃænd ruzi ke ʔindʒayim hitʃ ʔetefaqe qeyreʔadiʔi næyoftade	No	N	N3
7	شاید شما اعتماد به نفس خودتون رو از دست دادین. ʃayæd ʃoma ʔeʔtemadebenæfse xodetuno ʔæz dæst dadin	No	N	N4
8	آدم بهتره برای شروع هر کاری اول به همه جوانب فکر کنه بعد تصمیم بگیره. ʔadæm behtære bæraye ʃoruʔe hær kari ʔævvæl be hæmeye dʒævaneb fek kone bæʔd tæsmim begire	Yes	N	N2
9	خدایا من چقدر خوشبختم، باید منو دخترم افتخار کنیم. xodaya mæn tʃeqædr xoʃbæxtæm bayæd mæno doxtæræm ʔeftexar konim	Yes	H	H1
10	باریکلا، خیلی کار خوبی کردی. barikæla xeyli kare xubi kærdi	No	H	H3
11	راستی من چقدر از این بازی‌ها خوشم میاد، بازم بگو، تعریف کن. rasti mæn tʃeqædr ʔæz ʔin baziha xiʃæm miyad bazæm begu tæʔrif kon	No	H	H4
12	الهی که دورت بگردم ننه، الهی که عباس فدات بشه. ʔelahi ke doret begærdæm næne ʔelahi ke ʔæbbas fædat beʃe	Yes	H	H2
13	خدایا من قادر نیستم رنج محبوبم رو ببینم. xodavænda mæn qader nistæm rændʒe mæhbubæm ro bebinæm	No	S	S3
14	من همه چیز رو از دست دادم، یک تصویر داشتم که اونم دیگه ندارم. mæn hæme tʃiz ro ʔæz dæst dadæm yek tæsvir daʃtæm ke ʔunæm dige nædaræm	Yes	S	S1
15	خدایا، خداوندا، یعنی من باید تا ابد توی این قفس طلایی اسیر باشم؟ xodaya xodavænda yæʔni mæn bayæd ta ʔæbæd tu ʔin qæfæse tælayi ʔæsir baʃæm	Yes	S	S2
16	چیزی که با اون همه عشق شروع شده، با این همه نفرت تموم بشه؟ tʃizi ke ba ʔun hæme ʔeʃq ʃoruʔ ʃode ba ʔin hæme nefræt tæmum beʃe	No	S	S4

Column E is related to emotions (A: Angry, N: Neutral, S: Sad, H: Happy), column R is related to requirements (Yes, No), and column N is related to number

3.2 The process of collecting audio speech

The Sony F-VK98 microphone was used to collect the data in an acoustic room. Also, according to Tables 2 and 3, a sheet containing the texts was given to the speakers (There is a separate text for male and female speakers) so that the speakers could utter the sentences in target emotions. We carefully selected the sentences that created natural emotional scenarios in order to ensure authenticity in imitating the emotions almost naturally. In addition, in order to convey emotions better, we trained the speakers before starting the recordings. To achieve this, similar audios were played to arouse emotions in the speakers. Also, a speech expert was present in all recording sessions to retrain the speakers before starting each session ensuring that they could imitate the requested emotions well. It should be mentioned that three texts were considered for each speaker. Out of these three sentences, two sentences were mandatory and the speaker had the option of selecting the third sentence from two optional sentences to have a better imitated emotion by each speaker. Each speaker articulated mandatory (and optional) texts for the four emotion classes (Neutrality, Happiness, Anger, and Sadness) at least twice. Recordings were conducted on multiple sessions in different days to

Table 3 Sentences for female speakers

N	Sentences (for female speakers)	R	E	ID
1	مگه اینجا خونه خالهاست که هر وقت خواست بیاد و هر وقتم خواست بره. mæge ʔindʒa xuneye xalæst ke hær væxt xast biyado hær væxt hæm xast bere	Yes	A	A1
2	صبرکن بیام، یه عباس آقایی بسازم که هشت تا از بغلش در بیاد. عباس آقا، عباس آقا sæb kon biyad ye ʔæbbas ʔaqayi besazæm ke hæʃta ʔæz bæqæleʃ dær biyad ʔæbbas ʔaqa ʔæbbas ʔaqa	No	A	A3
3	لطفا امروز اصلاً دورو بر من نپلک چون اصلاً اصلاً حوصلتو ندارم. lotfæn ʔemruz ʔæslæn dorobære mæn næpelek tʃon ʔæslæne ʔæslæn hoselæto nædaræm	Yes	A	A4
4	حالا وایستادی اون گوشه با اون چشمای دریدهات زل زدی به ما که چی؟ hala vaysadi ʔun guʃe ba ʔun tʃeʃaye deridæt zol zædi be ma ke tʃi	No	A	A2
5	دوستان پدر، استاد بهار و علی اکبر خان دهخدا در خانهشان مهمانانند. dustane pedær ʔostad bæhar væ ʔæli ʔækbær xane dehxoda dær xaneyeʃan mihmanænd	No	N	N1
6	مرد جوان را که از حال رفته به داخل خانه میآورند و در پی دنبال طبیب میفرستند. mærde dʒævan ra ke ʔæz hal ræfte be daxele xane miʔaværænd væ dær peye tæbib miferestænd	No	N	N3
7	پروین اعتصامی از کودکی نزد پدر ادب دان و مادر شعر دوستش با سرودن مأنوس شد. pærvine ʔeʔtesami ʔæz kudæki næzde pedære ʔædæbdano madære ʃeʔrdustæʃ ba sorudæn mæʔnus ʃod	Yes	N	N4
8	برادرش ابولفضل در میان سایر برادرانش به اون نزدیک تر است. bæradæræʃ ʔæbolfæth dær miyane sayere bæradæranæʃ be ʔu næzdiktæræst	Yes	N	N2
9	انشالله به جای صد سال، صد و بیست سال زنده باشی. ʔinʃalla bedʒaye sæd sal sædobis sal zende baʃi	Yes	H	H1
10	شما انقدر به دختر من محبت کردین که تا عمر دارم محاله فراموش کنم. ʃoma ʔenqæd be doxtære mæn mohæbbæt kærdin ke ta ʔomr daræm mæhale færamuʃ konæm	No	H	H3
11	آه، خدایا شکرت که تو این دنیا به زنی برخوردم که این همه بامحبته. ʔah xodaya ʃokret ke tuye ʔin donya be zæni bærxordæm ke ʔin hæme bamohæbbæte	No	H	H4
12	خوش اومدین، بفرمایین، خواهش میکنم. xoʃ ʔumædin befærmayin xaheʃ mikonæm	Yes	H	H2
13	خدایا، چقدر بدبختیم، چی به سرمون اومده؟ چیکار باید بکنیم؟ xodaya tʃeqædr bædbæxtim tʃi be særemun ʔumæde tʃikar bayæd bokonim	Yes	S	S3
14	به هر دری زدم، هرکاری کردم اما کسی حاظر نشد به ما کمک کنه. be hær dæri zædæm hær kari kærdæm ʔæmma kæsi hazer næʃod be ma komæk kone	No	S	S1
15	کیه که بتونه یه سال تموم گوشهٔ خونه بشینه و درس بخونه؟ kiye ke betune ye sale tæmum guʃeye xune beʃine væ dærs bexune	No	S	S2
16	فقط به خاطر خودش و منفعتش با من دوست بوده، نه به خاطر خودم. fæqæt be xatere xodeʃo mænfeʔæteʃ ba mæn dus bude næ bexater xodæm	Yes	S	S4

Column E is related to emotions (A: Angry, N: Neutral, S: Sad, H: Happy), column R is related to requirements (Yes, No), and column N is related to number

Table 4 Mean and standard deviation of speakers' ages

	Mean	SD
Female	25.27	2.30
Male	25.60	1.36
All	25.38	2.13

minimize fatigue effects. All of these mentioned scenarios ensured authenticity in imitating the emotions for non-professional speakers.

3.3 Equipment and collection methods

The Sony F-VK98 microphone was used to collect the data in an acoustic room. In addition, the sampling rate was considered to be 44,100 Hz in order to preserve the quality of data. According to Sect. 3.2, the text was given to speakers so that they could practice properly articulating the phrases and expressing their emotions. Also, during the recording, in order to stimulate more emotions, similar audio files were played for speakers.

3.4 Information of speakers

All of the participants were native Persian speakers with a complete Persian accent. However, people were chosen from different cities in Iran in order to preserve the variety of ways that the Persian language is expressed in various locales. 28 speakers participated in the database recording (17 males and 11 females). The mean age of the speakers is 25.38 and the standard deviation (SD) of their age is 2.13. The information is tabulated in Table 4.

3.5 Number of utterances for each speaker

The number of utterances for each speaker is shown in Fig. 2. As can be seen in this figure, the highest number of recorded speeches belongs to the speaker M06 with 143 utterances and the lowest number belongs to speakers M01, M02, M07, M12, and M19 with 24 utterances.

After recording the utterances, 1614 audio data equivalent to 2 h and 18 min were collected. Among these audio files, 186 files were deleted due to low quality (because of background noise or recording problems).

3.6 Labeling process

Although the texts given to the speakers have labels, the speaker's emotion cannot be trusted to be the same as the desired emotion. Therefore, the utterances were relabeled by native speakers.

We have proposed a new labeling system which is more reliable and makes the recording activity faster unlike the existing articles. In previous works, the labeling process requires the presence of the voter in a specific place and use paper questionnaires for labeling. This is a very time-consuming and expensive task because in addition to the need for people to be present, the audio file must be played and the voter must comment on it. If they played audio file does not match the name of the file written in the questionnaire sheet, the voter will comment on the wrong audio files without being aware of the issue. Therefore, the possibility of mistakes in these methods is very high. However, we will modify these drawbacks by our suggested labeling method.

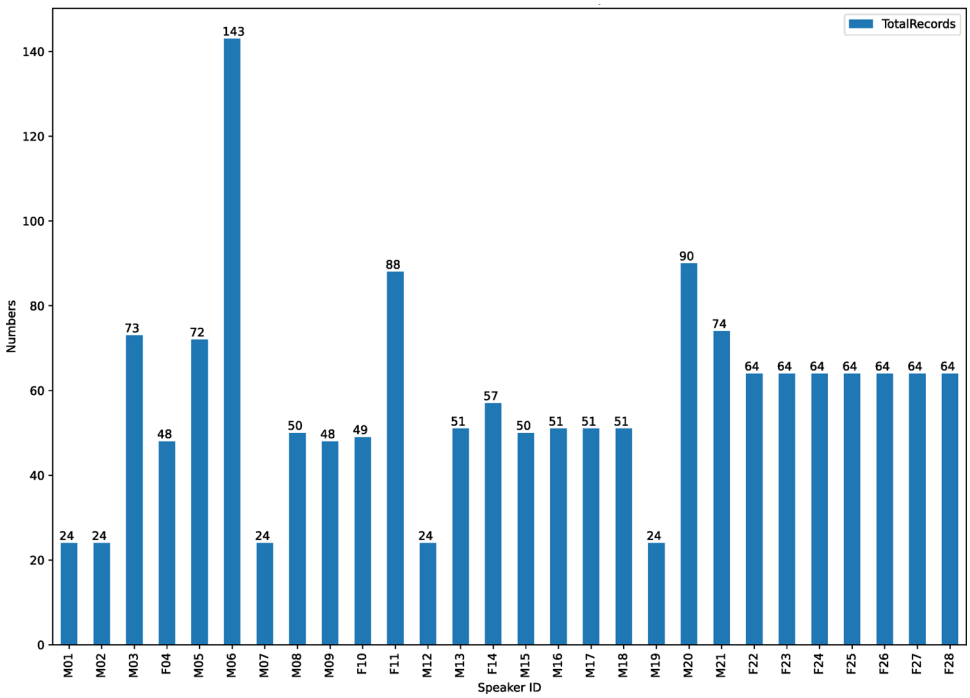


Fig. 2 The number of utterances of each speaker in AutESD dataset

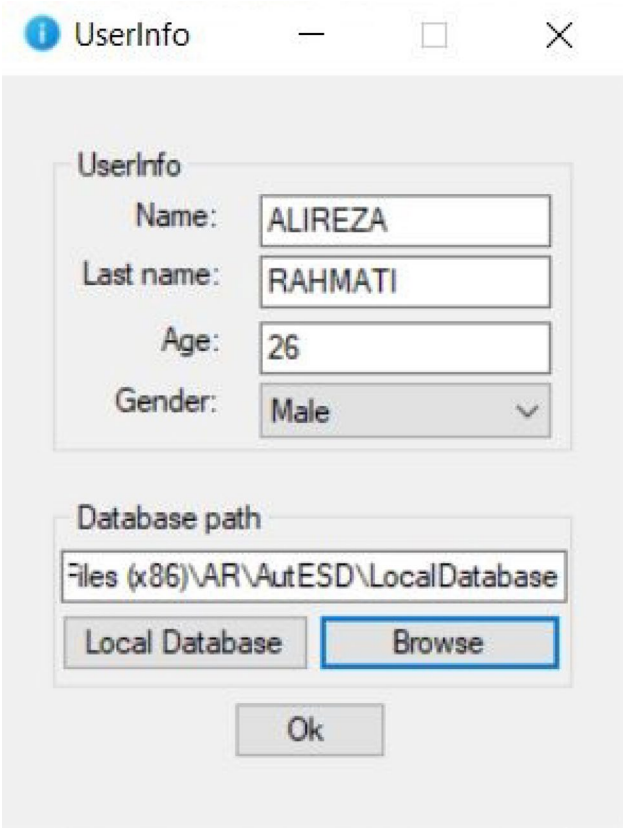


Fig. 3 The voter's initial entry and personal information registration

Unlike most datasets (according to IEMOCAP [34]), in this paper, a new labeling software is presented to reduce the existing challenges. In addition to reducing costs, voters can vote anytime and for any number of utterances they want. There is no issue with the played audio file not matching the original file name because the program saves the voter's comments in an Excel file based on the name of the audio file. Hence, this software saves money and time, and also enables us to use more people for labeling. We explain the details of this proposed labeling procedure in the following section which develops the labeling software.

3.6.1 Labeling software

According to Fig. 3, the users enter their information after launching the labeling program so that the software can use it to register the voter's information. As can be seen in Fig. 4, after entering the program, the user has the option to vote for one of the four emotions, as well as facilities for playing and turning off the audio.

For each audio file, there are two "strong" and "weak" emotional states, each of which contains "happy", "neutral", "sad", "angry", and "none" options. The "none" option is meant unambiguous emotion which could be hardly understood or an emotion outside the category of the existing ones (Fig. 5). The user must select an option from the category of strong emotions in order to register a vote and listen to the

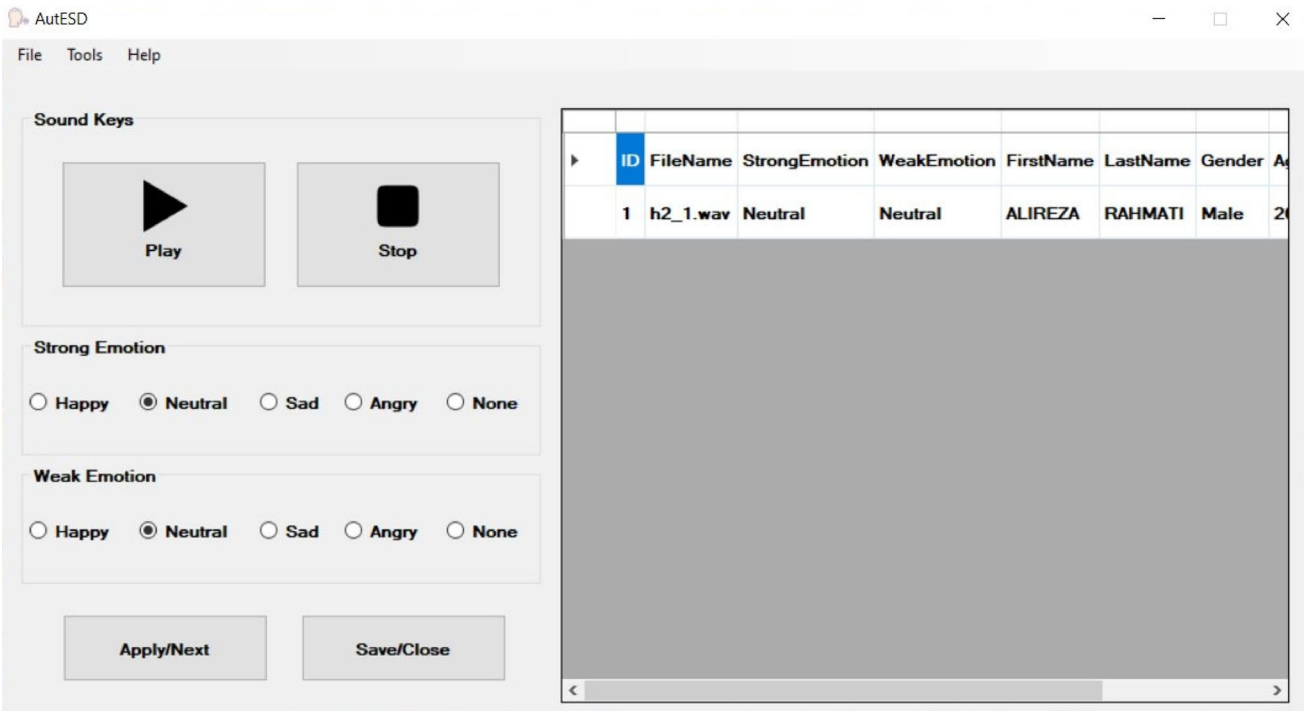


Fig. 4 The AutESD software interface for emotional speech labeling. This tool allows annotators to assign two emotion labels (strong and weak) to each speech sample simultaneously. Providing a strong emotion label is mandatory to ensure that no utterance is left unlabeled, while the "None" option can be used when the annotator cannot identify an emotion

ID	FileName	StrongEmotion	WeakEmotion	FirstName	LastName	Gender	Age	Date
1	M15S1R2D2.wav	Sad	Neutral	ALIREZA	RAHMATI	Male	26	19/09/2022 15:19:28
2	M18H2R2D3.wav	Happy	Neutral	ALIREZA	RAHMATI	Male	26	19/09/2022 15:20:55
3	F04H1R2D1.wav	Happy	Neutral	ALIREZA	RAHMATI	Male	26	19/09/2022 15:21:03
4	F11N1R1D1.wav	Neutral	Neutral	ALIREZA	RAHMATI	Male	26	19/09/2022 15:21:11
5	M05H1R1D4.wav	Neutral	Neutral	ALIREZA	RAHMATI	Male	26	19/09/2022 15:21:24
6	M03H4R2D3.wav	None	None	ALIREZA	RAHMATI	Male	26	19/09/2022 15:21:53
7	M21S3R1D6.wav	Sad	Sad	ALIREZA	RAHMATI	Male	26	19/09/2022 15:22:00
8	F11A3R2D3.wav	Angry	Angry	ALIREZA	RAHMATI	Male	26	19/09/2022 16:22:25
9	F10S1R2D4.wav	Sad	Sad	ALIREZA	RAHMATI	Male	26	19/09/2022 16:22:31
10	M03H3R1D4.wav	Neutral	Neutral	ALIREZA	RAHMATI	Male	26	19/09/2022 16:22:36
11	M15A3R2D5.wav	Angry	Angry	ALIREZA	RAHMATI	Male	26	19/09/2022 16:22:42
12	M03H1R1D4.wav	Neutral	Happy	ALIREZA	RAHMATI	Male	26	19/09/2022 16:22:51

Fig. 5 An example of a file saved by the user with our proposed and developed labeling software

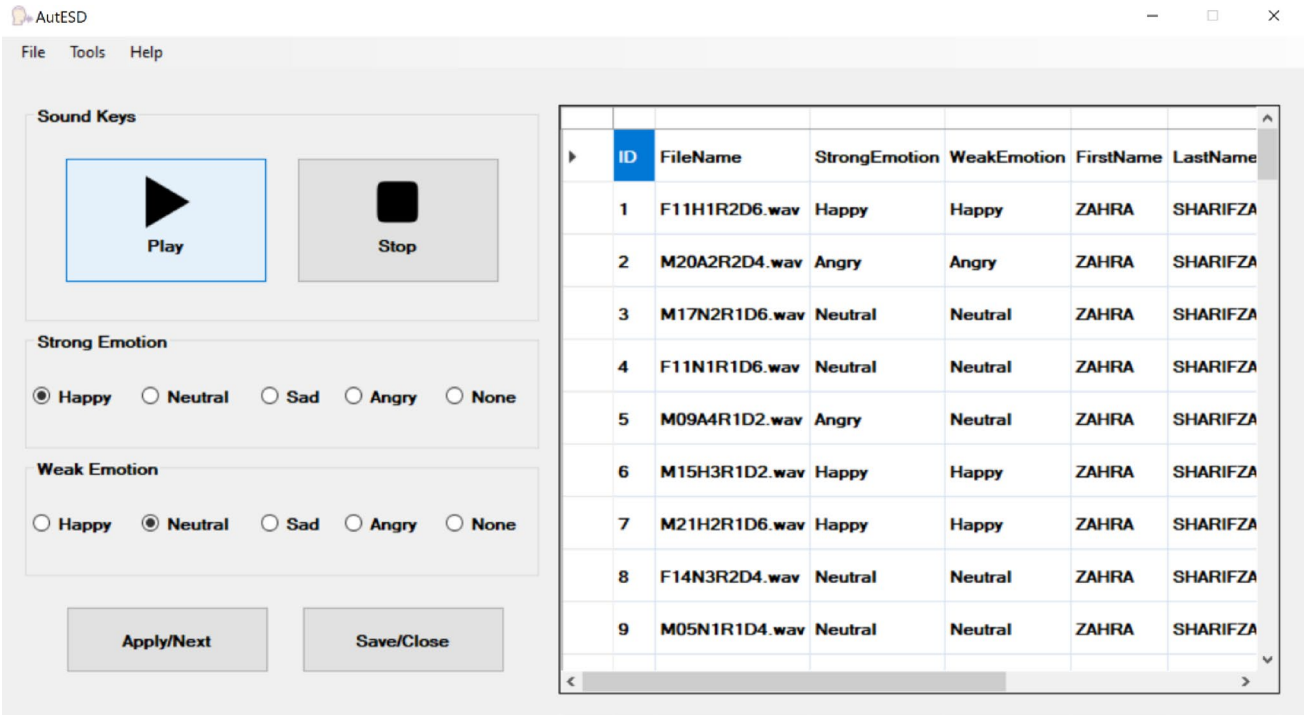


Fig. 6 The AutESD software interface for emotional speech labeling for another user showing both strong and weak labels

next utterance. After selecting the desired option, the user can proceed to the next audio file. However, according to Figs. 4 and 6, there is also a section for weak emotions so that voters can mark a weaker emotion as a second priority. It is worth mentioning that the volunteers for this voting process are not aware of the true file label. To determine the final ground-truth label for each utterance, we applied a two-stage majority rule:

1. Strong-level majority:
   If one emotion class received the majority of Strong votes, that class was assigned as the final label.
2. Fallback to Weak:
   If Strong votes were insufficient or tied across classes, we examined the Weak-level votes. A majority at this level determined the final label.
3. Rejection rule:
   If no majority emerged at either level—i.e., ties or too few votes—the utterance was marked as ambiguous and removed from the final dataset.

The software includes built-in mechanisms to prevent duplicate labeling and automatically save annotations, ensuring that each sample is uniquely labeled and reducing the risk of data loss. Finally, as shown in Figs. 5 and 7, users can save the results from the tools section in CSV or ZIP formats, which provides well-organized outputs containing

	0	1	2	3	4	5
0	ID	FileName	StrongEmotion	WeakEmotion	Gender	Age
1	1	F11H1R2D6.wav	Happy	Happy	Female	25
2	2	M20A2R2D4.wav	Angry	Angry	Female	25
3	3	M17N2R1D6.wav	Neutral	Neutral	Female	25
4	4	F11N1R1D6.wav	Neutral	Neutral	Female	25
5	5	M09A4R1D2.wav	Angry	Neutral	Female	25
6	6	M15H3R1D2.wav	Happy	Happy	Female	25
7	7	M21H2R1D6.wav	Happy	Happy	Female	25
8	8	F14N3R2D4.wav	Neutral	Neutral	Female	25
9	9	M05N1R1D4.wav	Neutral	Neutral	Female	25
10	10	M06H1R1D6.wav	Happy	Happy	Female	25
11	11	M17A3R1D6.wav	Sad	Neutral	Female	25
12	12	M21S1R1D4.wav	Sad	Neutral	Female	25
13	13	M21A2R1D4.wav	Angry	Angry	Female	25
14	14	M03A3R2D4.wav	Angry	Angry	Female	25

Fig. 7 An example of another file saved by the user in Fig. 6 with our proposed and developed labeling software

information such as the audio file name, selected emotions, voter's name, gender, age, and the voting date.

This labeling software also supports multi-user and remote participation, enabling the involvement of more annotators, increasing dataset size, diversity, and annotation accuracy compared to previous methods, while reducing the overall cost of data collection.

3.6.2 The final result of labeling

Labeling was done by 16 people (6 male and 10 female) In order to achieve the best accuracy, these people were selected from professional or semi-professional students of the speech laboratory in the doctoral or master's degree. We omit the file in these cases: (1) when the number of votes in two or more classes are equal while they have the highest votes among classes, (2) when None gets the highest votes (Showcasing the emotion is not detectable). (3) When the voting accuracy is low in some votes results which might be caused due to non-professional speakers for imitating emotions. Therefore, 186 audio files were deleted. remaining 1428 audio files. The labels were just changed in some of these files, because the votes results showed another emotion (For example the sentences must be acted happily but the speaker acted them angrily and voting revealed the true emotions). Figure 8 depicts the final number of each speaker's utterances. The minimum number of audio files is related to speaker number M02 with 2 utterances and the maximum is related to speaker number M06 with 129 audio files.

The certainty of the detected emotions for utterances can be observed in percentages in Fig. 9. According to this figure, 1244 utterances have been labeled with more than 95% accuracy. From the 1614 recorded speeches, 1428 files remain. Figure 9 shows that the majority of the utterances (1244 utterances) have a labeling accuracy of over 95%, which means that the recorded speech has a strong feeling.

As shown in Fig. 10, the results are divided into four classes of emotions: happiness, anger, sadness, and neutrality which include 365 utterances for anger, 353 utterances for happiness, 346 utterances for sadness, and 364 utterances for neutrality. This makes our AutESD dataset an almost balanced corpus having approximately similar number of utterances in each emotion class.

Furthermore, we have examined the finalized labeled files in terms of speaking time, as shown in Table 5. This table displays the average speech duration per text, the number of speeches per text, and the total speech duration per text. (refer to the ID column in Tables 2 and 3 for the corresponding text identifiers).

Table 6 summarizes the inter-rater agreement for the AutESD dataset based on Fleiss' Kappa [36, 37] across varying numbers of annotators. For groups with two or more voters, Kappa values ranged from 0.70 to 1.00, indicating substantial to almost perfect agreement according to Landis and Koch's interpretation scale. Specifically, strong-emotion samples yielded slightly higher agreement (average $\kappa \approx 0.84$) compared to weak-emotion samples (average $\kappa \approx 0.78$), confirming consistent labeling behavior across different group sizes and demonstrating the reliability and robustness of the annotation process.

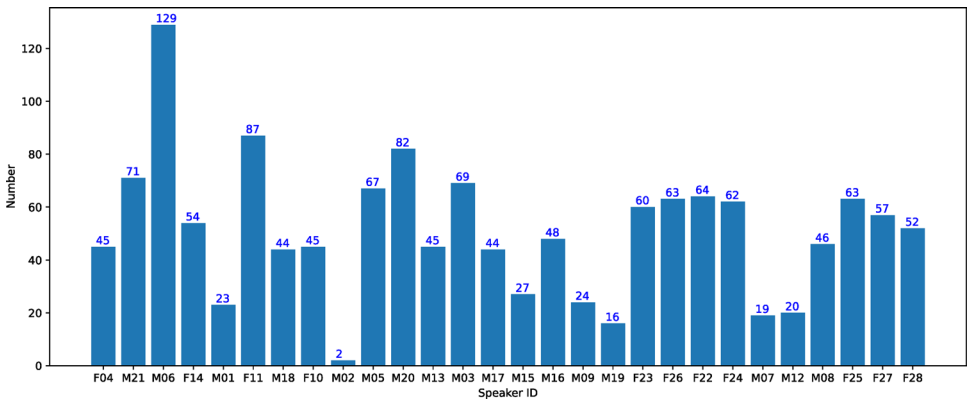


Fig. 8 The final number of each speaker's utterance in AutESD after running the proposed and developed labeling software

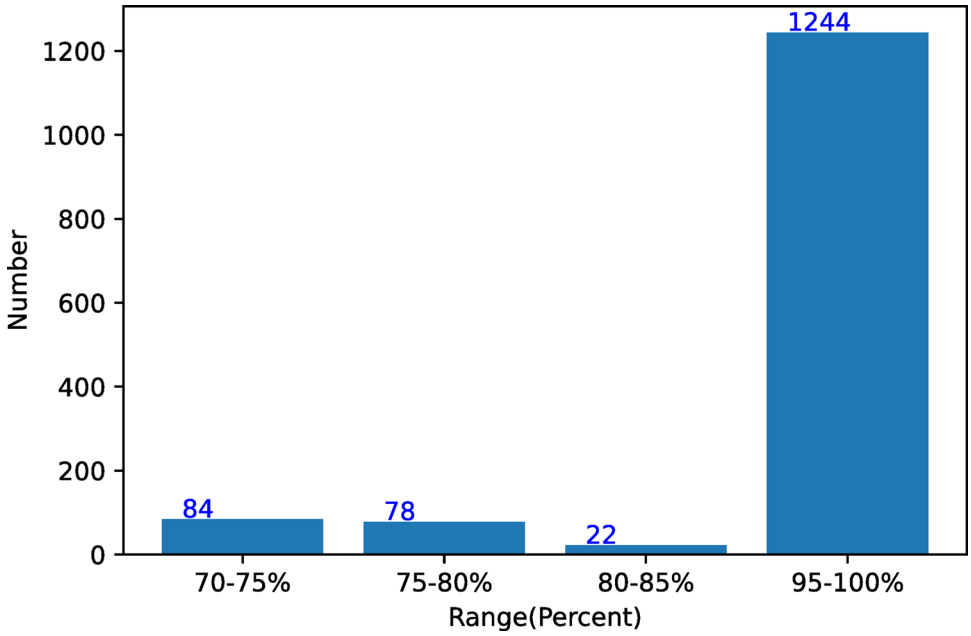


Fig. 9 Accuracy of labeling for audio files of AutESD in percentage. For example, 84 audio files have an emotion accuracy of 70–75%

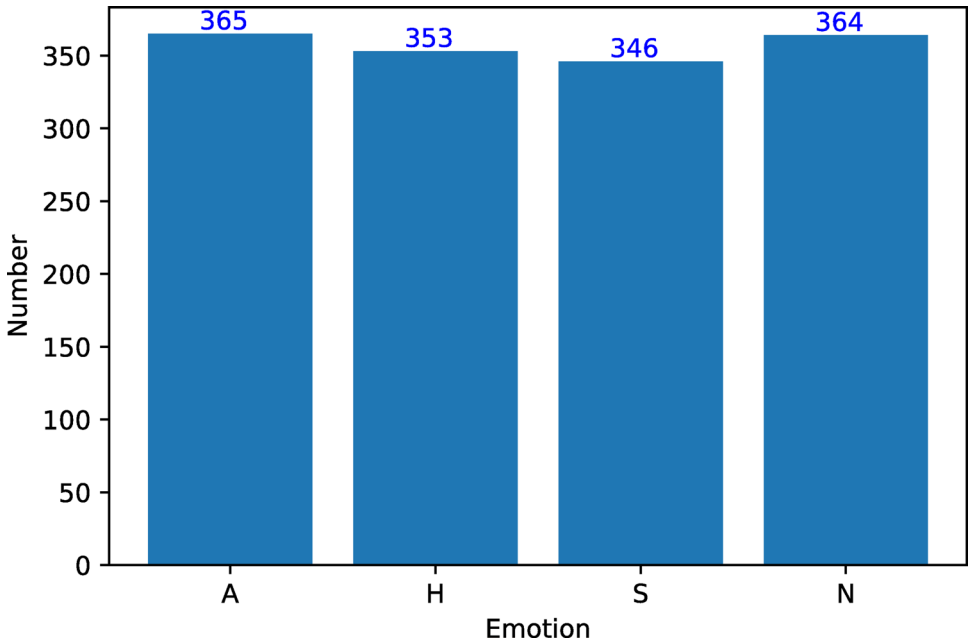


Fig. 10 The final number of utterances in each category of emotions in AutESD dataset

Previous emotional speech datasets reported varying inter-annotator agreement: Shemo [16] achieved a Cohen's Kappa of 0.64 (substantial agreement), EmoHopeSpeech [36] reported Fleiss' Kappa values between 0.75 and 0.85 for both Arabic and English, and EmoTa [37] obtained a Fleiss' Kappa of 0.74. In comparison, the AutESD dataset shows slightly higher agreement, with an average Fleiss' Kappa of 0.84, highlighting its strong annotation consistency relative to existing datasets.

3.6.3 Summary of autesd information

Table 7 provides a concise summary of the AutESD dataset, consolidating key information such as the number of speakers, utterances per emotion class, gender distribution, average age, recording conditions, and labeling methodology. This summary organizes details previously mentioned in multiple sections, giving readers a clear overview of the dataset.

Table 5 Average speaking time, number of speeches, and total speaking time based on the text

Text category	Average (Sec.)	Number of utterances	Sum of duration (Sec.)
A1	4.53	105	476
A2	5.15	118	607
A3	5.42	74	401
A4	4.06	68	312
H1	4.52	105	474
H2	4.17	105	437
H3	4.25	98	416
H4	5.39	45	242
N1	6.08	112	680
N2	5.63	109	613
N3	5.21	71	369
N4	5.06	72	364
S1	5.8	107	621
S2	5.92	104	616
S3	5.49	80	439
S4	5.49	55	290
Total	-	1428	7357 (2 H:2 M:37 S)

Table 6 Fleiss' kappa values for inter-rater agreement across different numbers of annotators in the autesd dataset for strong and weak emotion categories

Analyze	Fleiss' Kappa	
No. Voters	Strong	Weak
1 Voter	Not calculable	Not calculable
2 Voters	0.911	0.876
3 Voters	0.807	0.765
4 Voters	0.793	0.757
5 Voters	0.784	0.739
6 Voters	1.000	Not calculable (Only one item)

Table 7 Summary of AutESD dataset properties

Attribute	Value
Dataset Name	AutESD
Total Duration	2 h 2 min
No. of Utterances	1,428
No. of Speakers	28 (17 M, 11 F)
Age Range	23–30 (Mean = 25.4, SD = 2.1)
Emotion Classes	Neutral (N), Happy (H), Angry (A), Sad (S)
Dataset Type	Semi-natural emotional speech
Sampling Rate	44.1 kHz, 16-bit
Utterance Duration	4–6 s
Recording Environment	Acoustic room at Amirkabir University of Technology (AUT)
No. of Utterances per Class	365 (A), 353 (H), 346 (S), 364 (N)
Labeling Method	Two-level (Strong/Weak) with majority voting (the proposed software in this paper)
No. of Removed Files	186 (due to low confidence or unclear emotions)

3.6.4 Naming data

The data are named according to Fig. 11; Table 8. The first character of the name is related to the speaker's gender, the second and third characters represent the speaker's ID, while the fourth and fifth characters represent the emotion of speech and text number. The recording number is represented by the sixth and seventh characters, which range from one to a maximum of three recordings. The recording day is represented by the eighth to tenth characters (1st day to 11th day).

4 The preview of some data

Figures 12 and 13 show the waveforms and logarithmic Mel spectrograms for four anger, neutrality, happiness, and sadness categories of emotions after data collection and final labeling of utterances.

5 Dataset evaluation

In this section, the collected dataset is evaluated for speech emotion recognition task using three models of Support Vector Machine (SVM), K-Nearest Neighbor (K-NN), and Deep Neural Network (DNN).

5.1 Data preprocessing

As a crucial element for converting raw signals to high-level features, the Mel-Frequency Cepstral Coefficients (MFCC) with 20 coefficients are extracted from all files. The windowing length is set to be 25ms with a 15ms time shift.

5.2 Models configuration

We assessed the collected dataset by employing the SVM, KNN, DNN (CNN-based), and a transformer model utilizing a 5-fold cross-validation approach. To this end, we used a speaker independent case for train and test folds, i.e. the speakers used for training the models are different from those for testing in each fold. This is a stronger condition than speaker dependent case which uses the same speakers in train and test sets. Moreover, we have also used a speaker-dependent scenario for the SVM and KNN models for comparison. Specifically, the Radial Basis Function (RBF) was employed as the kernel for the SVM analysis. Additionally, for KNN, the number of neighbors (k) is set to 5, and DNN model is set according to Table 9. We have run all models on a PC with Corei7 CPU and RAM of 24GB. The GPU of this computer is GTX960m having 2GB RAM.

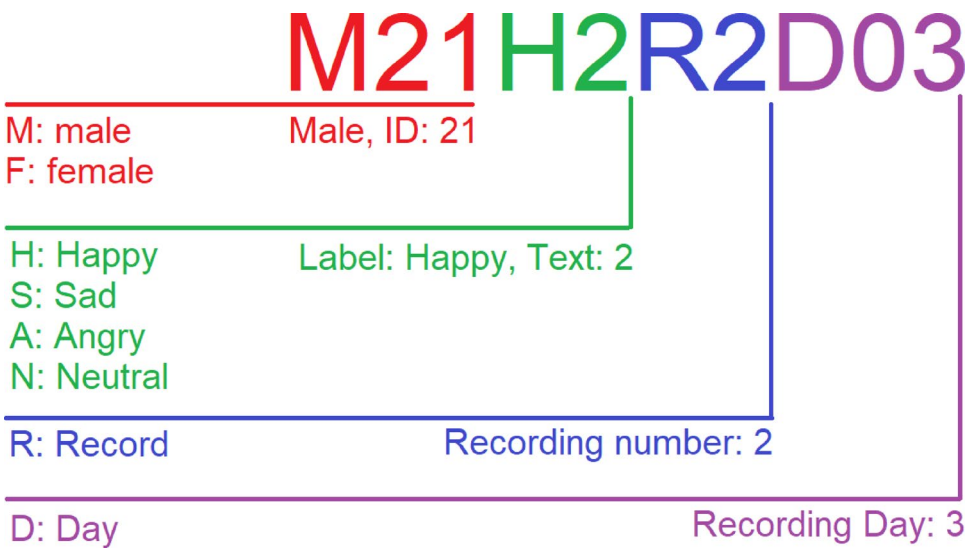


Fig. 11 Illustration of the labeling and file-naming convention in the AutESD dataset. Each code encodes the speaker's gender and ID (M/F and number), emotion label and text number (H/S/A/N and number), recording number (R), and recording day (D). For example, M21H2R2D03 represents male speaker 21, utterance 2 labeled as happy, recorded in session 2 on day 3

Table 8 Data naming method for the files of autesd

Character number	Meaning	Details
1	Speaker's gender	M: Male F: Female
2, 3	Speaker's ID number	0:28
4	The emotion that was conveyed from the utterance	H: Happiness S: Sadness A: Anger N: Neutrality
5	Sentence's number	1:4
6, 7	Recording's number	R1-R3
8, 9, 10	Speech recording day	D01:D11

5.3 Evaluation

5.3.1 SVM

According to the confusion matrix in Fig. 14, the highest and lowest accuracies on the test dataset are observed in the Sad and Neutral classes, with accuracies of 0.75 and 0.47, respectively. Moreover, the most misclassification is overally related to Neutral class, while according to Fig. 14, Anger is mostly confused by Happiness which is natural based on their high arousal values. As indicated in Table 10, the SVM model displayed the unweighted average recall (UAR) of 55.1±3.3% on the test dataset in the speaker-independent scenario. We also assessed the performance of AutESD using SVM model in the speaker-dependent scenario shown in Table 11. According to Table 11, the SVM

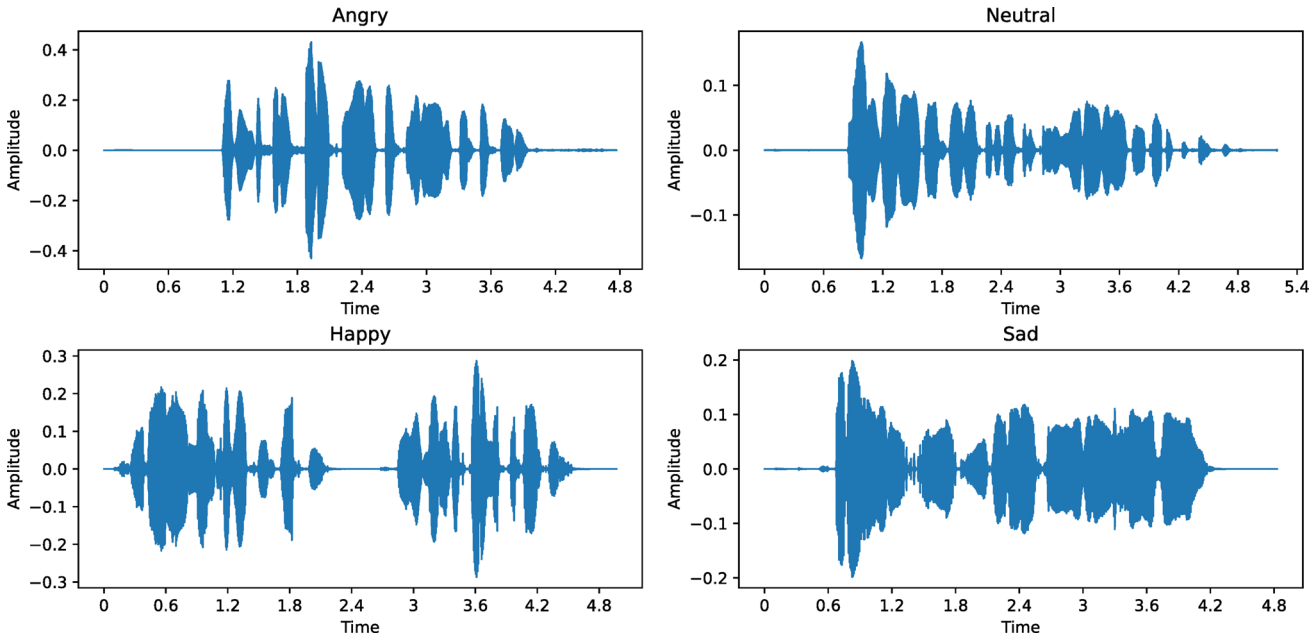


Fig. 12 Waveforms of four audio files with different emotions in AutESD dataset

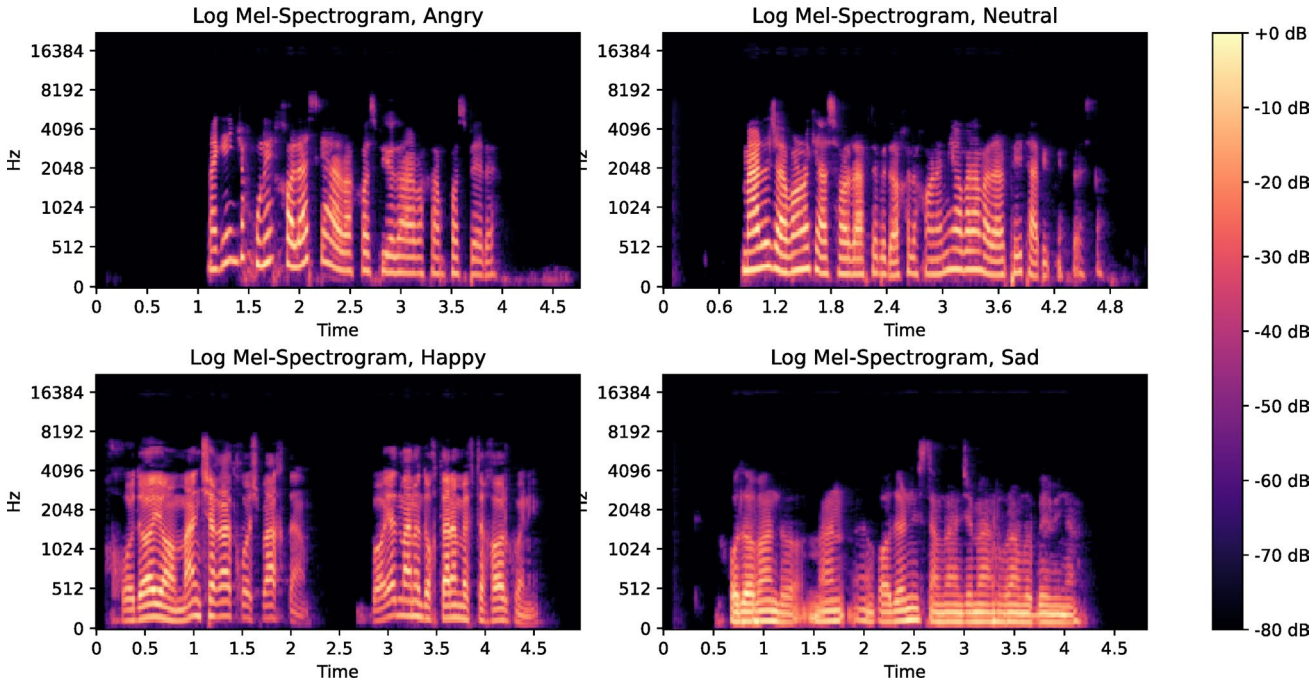


Fig. 13 The logarithmic Mel-spectrogram of four audio files with different emotions in AutESD dataset

Table 9 DNN layer configuration

Block name	Layers	Settings
CNN Block 1	CNN	NK = 16, KS=(3,5), ST=(1,2)
	ReLU activation	-
	Max polling	KS=(3,5), ST=(3,5)
CNN Block 2	CNN	NK = 32, KS=(3,3), ST=(1,1)
	ReLU activation	-
	Max polling	KS=(1,2), ST=(1,2)
CNN Block 3	CNN	NK = 64, KS=(3,3), ST=(1,1)
	ReLU activation	-
	Max polling	KS=(1,2), ST=(1,2)
Linear Block 1	Linear	64 Neurons
	ReLU activation	-
Linear Bock 2	Linear	16 Neurons
	ReLU activation	
Linear Block 3	Linear	4 Neurons
	Softmax	-

NK: number of kernels, KS: kernel size, ST: Stride

model displayed the unweighted average recall (UAR) of 56.3 ± 0.8% on the test dataset in the speaker-independent scenario. As expected, the speaker-independent scenario achieves lower accuracies due to its strong condition for selecting different speakers in train and test sets.

5.3.2 KNN

As shown in the confusion matrix in Fig. 15, the K-Nearest Neighbors (KNN) model exhibited varying accuracies on the test dataset, with the highest and lowest values observed in the Sad and Neutral classes, measuring 0.69 and 0.23, respectively. Notably, the Neutral class showed the highest misclassification rate on average, while Anger is mostly confused by Happiness as seen in natural emotions too. Table 12 illustrates the performance of the KNN model, indicating its UAR of 43.7 ± 1.8% on the test dataset in the speaker-independent scenario. Table 13 depicts the performance of the KNN model, indicating its UAR of 50.6 ± 3.7% on the test dataset in the speaker-dependent scenario, which is higher than speaker-independent one as expected.

5.3.3 DNN (CNN-based)

In an effort to improve the classification accuracy of the AutESD system, we explored a Deep Neural Network (DNN) architecture according to Fig. 16, made up of Convolutional Neural Network (CNN) blocks. The proposed CNN consists of three convolutional layers with an increasing number of features—16, 32, and 64, respectively—aiming to extract hierarchical features from the input data. These layers are followed by three fully connected layers with 64, 16, and 4 units, each activated by the Rectified Linear Unit (ReLU) function to introduce non-linearity to the model (the last activation is SoftMax to generate class probabilities), which is essential for capturing complex patterns. This model has 246,103 parameters in total.

The model is trained on a dataset split into training, validation, and testing sets (5-Fold with 70%, 10% and 20% respectively). As mentioned before, we have used the speaker independent setting for train and test splits in each

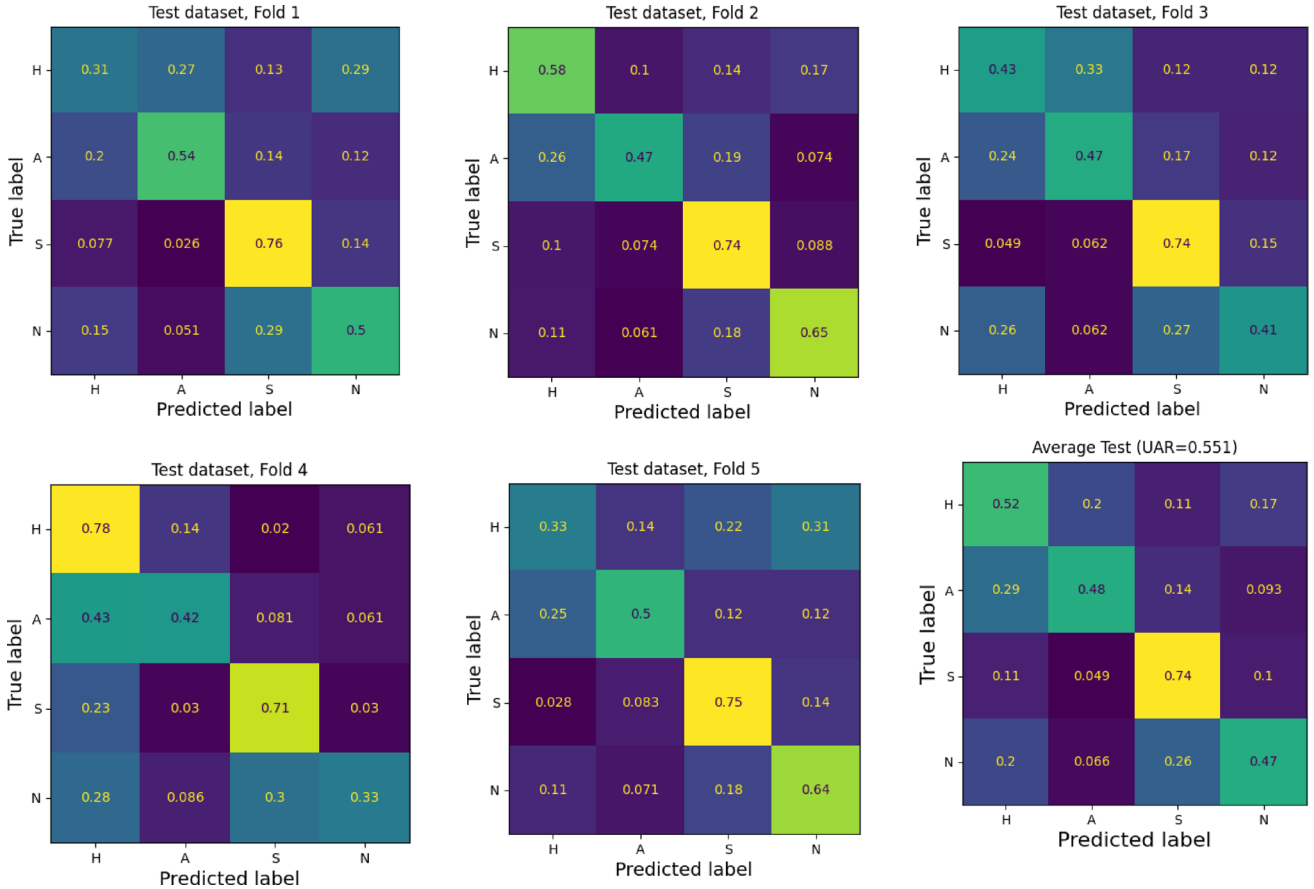


Fig. 14 Confusion matrices obtained from testing the SVM model on each of the 5 cross-validation folds, along with the matrix representing the average performance across all folds

Table 10 Evaluation of the autesd dataset using the SVM model in five folds in terms of UAR in the speaker-independent scenario using MFCC

	Train (UAR)	Test (UAR)
Fold 1	0.740	0.526
Fold-2	0.723	0.609
Fold 3	0.738	0.514
Fold 4	0.753	0.562
Fold 5	0.745	0.557
Average	0.739	0.551
Standard deviation	0	0.033

Table 11 Evaluation of the autesd dataset using the SVM model in five folds in terms of UAR in the speaker-dependent scenario using MFCC

	Train (UAR)	Test (UAR)
Fold 1	0.731	0.577
Fold-2	0.735	0.563
Fold 3	0.742	0.566
Fold 4	0.747	0.554
Fold 5	0.744	0.554
Average	0.740	0.563
Standard deviation	0.005	0.008

fold. The network is optimized using SGD optimizer with a categorical cross-entropy loss function. We tune the hyper-parameters based on the validation set performance, with a learning rate of 1e-3, batch size of 64, and 200 epochs of training.

Validation of the model is conducted by comparing predicted labels against true labels, using a confusion matrix and accuracy as performance metrics. Figure 17 shows the confusion matrices of DNN model for MFCC features. As illustrated in Fig. 17, the highest and lowest accuracies on the test dataset are observed in the Sad and Angry classes, with accuracies of 0.83 and 0.76, respectively. Moreover, the most misclassification is related to Angry class, while according to Fig. 17, Anger is mostly confused by Happiness which is natural based on their high arousal values. The model achieved an average unweighted testing accuracy of 76.4% for Mel-Spectrogram features and an average unweighted test accuracy of 78.9% using MFCC features.

Table 14 presents the test performance of DNN (CNN-based) model with both MFCC and Mel-spectrogram features in different terms of precision, recall, F1-score, and accuracy in the speaker-independent scenario. As shown

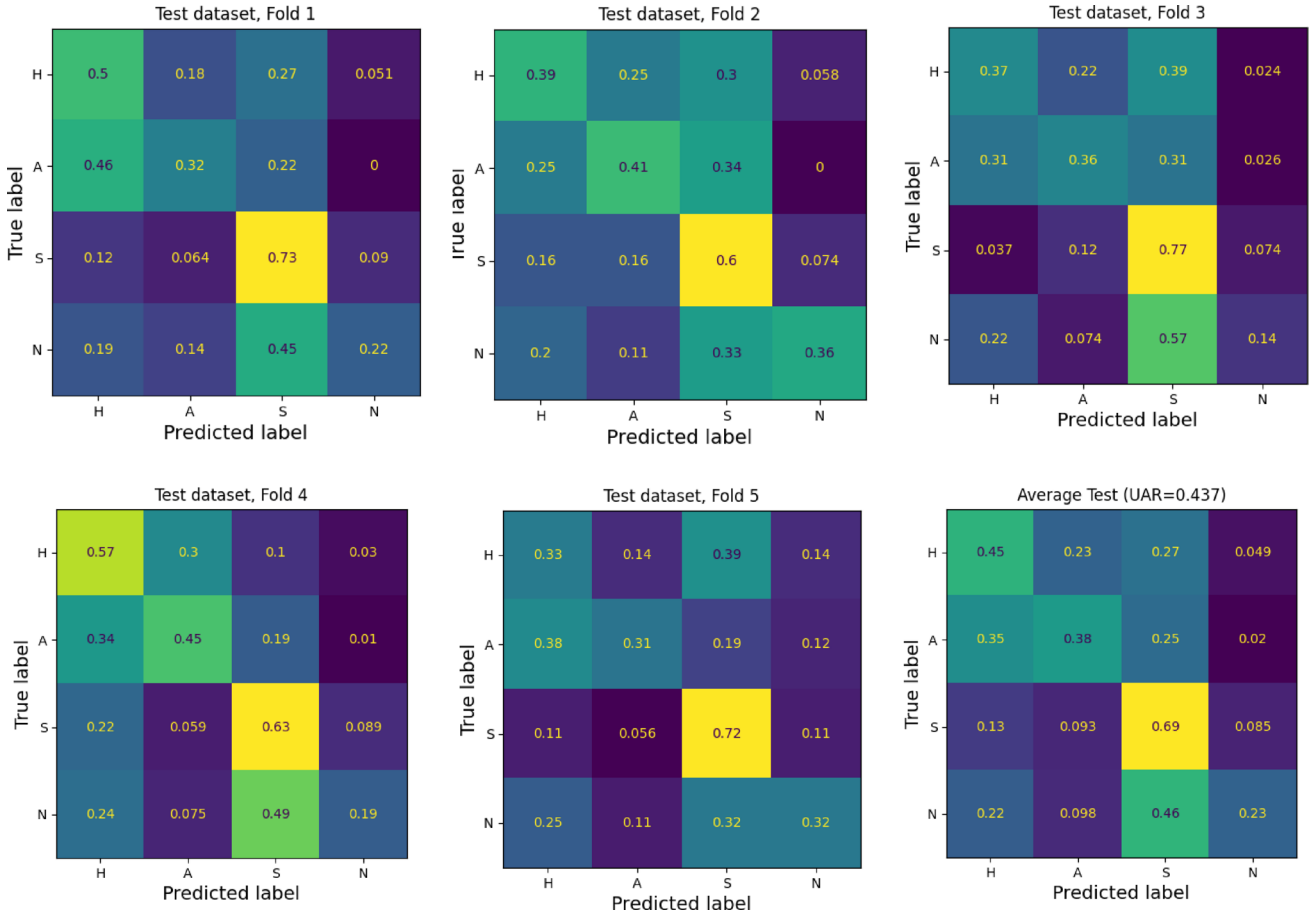


Fig. 15 Confusion matrices obtained from testing the KNN model on each of the 5 cross-validation folds, along with the matrix representing the average performance across all folds

Table 12 Evaluation of the autesd dataset using the KNN model in five folds in terms of UAR in the speaker-independent scenario using MFCC

	Train	Test
Fold 1	0.658	0.441
Fold 2	0.648	0.442
Fold 3	0.661	0.408
Fold 4	0.636	0.462
Fold 5	0.665	0.422
Average	0.653	0.437
Standard deviation	0.01	0.018

Table 13 Evaluation of the autesd dataset using the KNN model in five folds in terms of UAR in the speaker-dependent scenario using MFCC

	Train	Test
Fold 1	0.660	0.521
Fold 2	0.646	0.556
Fold 3	0.663	0.455
Fold 4	0.657	0.470
Fold 5	0.674	0.526
Average	0.660	0.506
Standard deviation	0.009	0.037

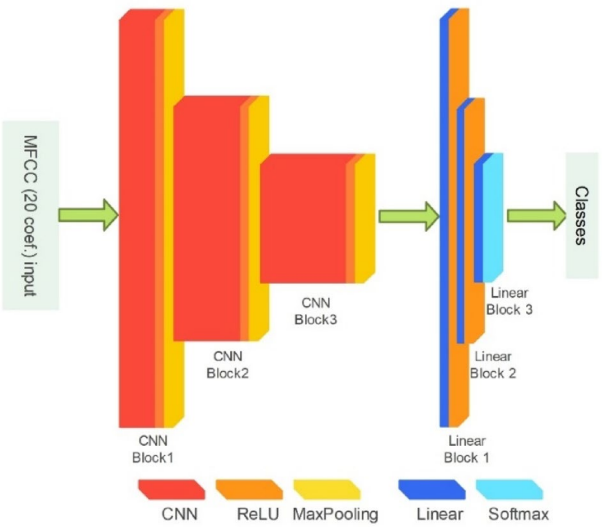


Fig. 16 DNN model structure. It consists of 3 CNN block and 3 Dense layer that are fed by MFCC features

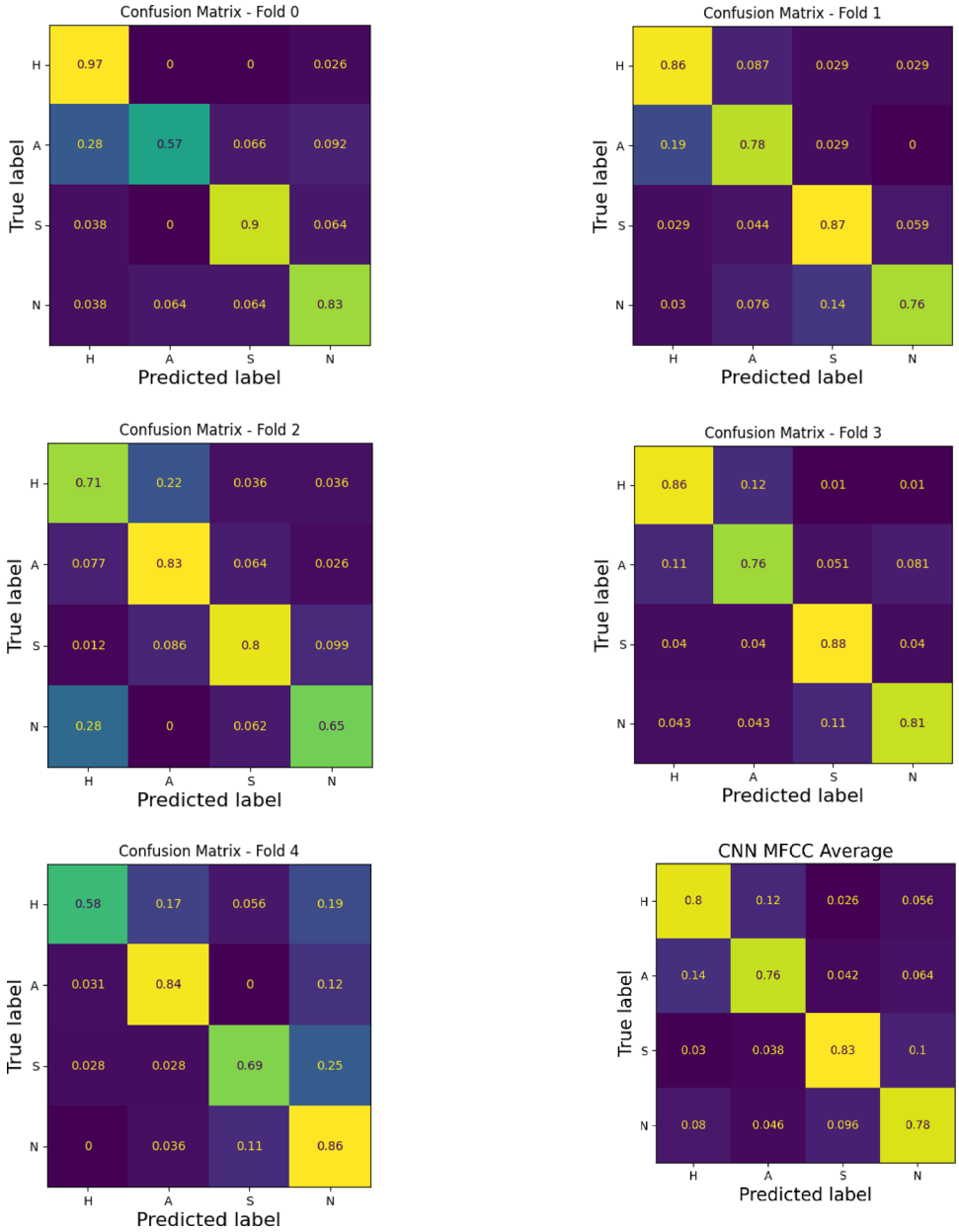


Fig. 17 Confusion matrices obtained from testing the DNN model on each of the 5 cross-validation folds along with its average using MFCC features

Table 14 Evaluation of the AutESD dataset using the DNN (CNN-based) model in five folds in the speaker-independent scenario

	Test (MFCC)		Test (Mel spec)	
	Mean	STD	Mean	STD
Precision	0.8002	0.032	0.7722	0.025
Recall	0.7908	0.035	0.7660	0.015
F1-score	0.7878	0.037	0.7616	0.015
Accuracy	0.7890	0.038	0.7640	0.017

in Table 14, the CNN model performed well on both feature types, with MFCC-based inputs yielding slightly better results (F1 = 0.7878, Accuracy = 0.789) than Mel-spectrograms (F1 = 0.7616, Accuracy = 0.764). The low standard deviations indicate stable performance across folds. These results suggest that MFCC features are more suitable for CNN architectures, mostly due to their compact representation of speech-relevant information.

5.3.4 Transformer

Figure 18 illustrates the block diagram we used for a transformer model. This model has 1,054,276 parameters in total. Table 15 presents the test performance of the transformer model with both MFCC and Mel-spectrogram features in different terms of precision, recall, F1-score, and accuracy in the speaker-independent scenario. According to Table 15, the Transformer achieved comparable results for both features, with MFCCs giving slightly higher accuracy (0.8329) but greater variability across folds (STD = 0.175).

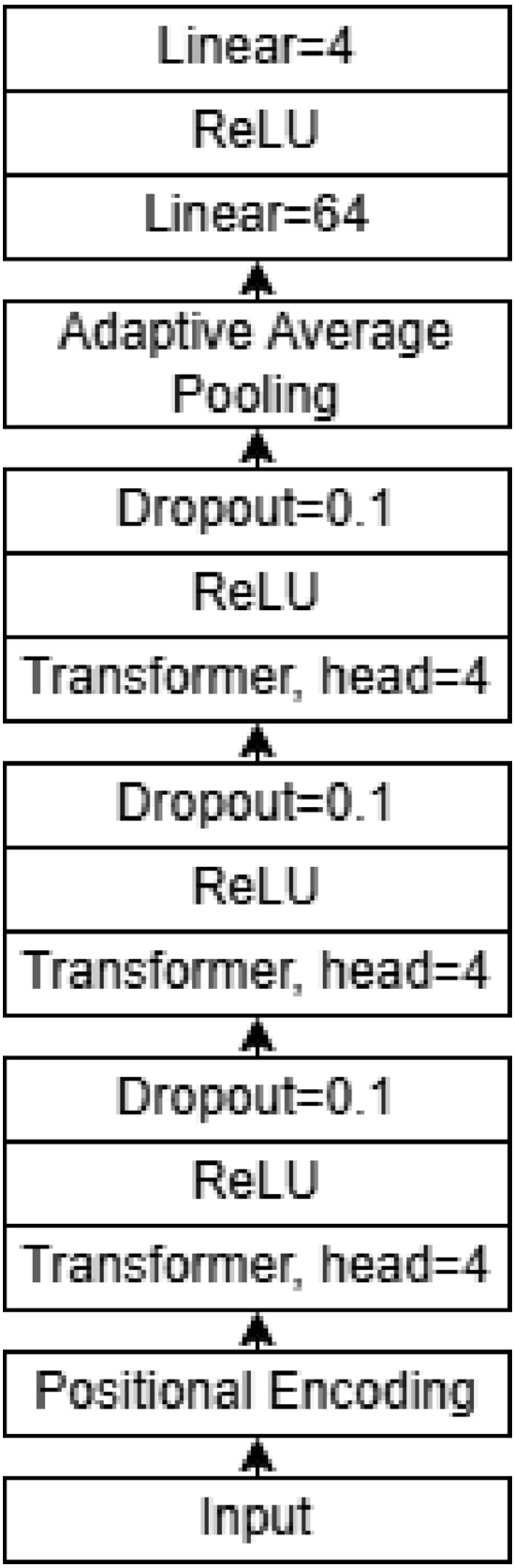


Fig. 18 The transformer model structure

Table 15 Evaluation of the AutESD dataset using the Transformer model in five folds in the speaker-independent scenario

	Test (MFCC)		Test (Mel spec)	
	Mean	STD	Mean	STD
Precision	0.7532	0.027	0.7623	0.039
Recall	0.7458	0.023	0.7530	0.044
F1-score	0.7434	0.021	0.7480	0.046
Accuracy	0.8329	0.175	0.7510	0.044

Mel-spectrograms accuracy is 0.751. The average recall for MFCC and Mel-spectrogram features are 0.7458 and 0.753, respectively.

5.3.5 Overall results

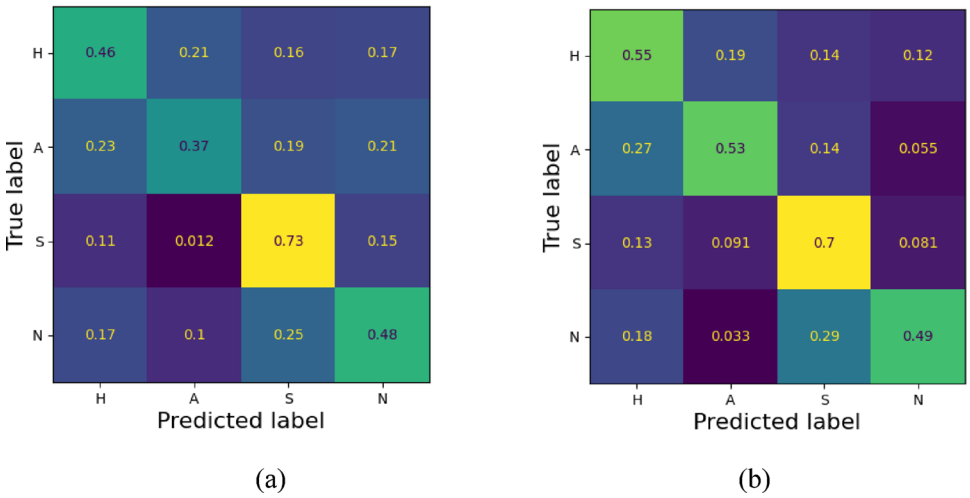
Table 16 compares the performance of different models on the AutESD dataset with results reported on the Shemo dataset [16] to have an approximate comparison, because we have used different model and fold settings. Therefore, assuming that we have used a speaker independent scenario in our train and test fold settings which is a strong condition, we could conclude that the models trained on AutESD achieved almost similar (or even higher) Unweighted Average Recall (UAR) compared to those evaluated on Shemo (using speaker-dependent scenario) on SVM and KNN models. The comparison between the results of Tables 10 and 11, as well as those in Tables 12 and 13 proves this claim. Moreover, we changed the settings to include separate UARs for both Females and Males to study the gender effects. Therefore, the reported results in Table 16 correspond to a gender-dependent scenario using the same gender in train and test sets. The low difference between male and female performance almost in all methods using either MFCC or Mel-spectrogram features suggest a low obtained bias in this subgroup.

Furthermore, the deep learning approaches yielded noticeably stronger performance according to Tables 14 and 15: the CNN achieved mean UARs of 0.79 (MFCC) and 0.766 (Mel-spectrogram), while the Transformer reached at 0.75 (MFCC) and 0.75 (Mel-spectrogram). These results suggest that the AutESD dataset enables more accurate emotion recognition, particularly when using advanced architectures and informative acoustic features.

Figure 19 depicts the confusion matrices obtained with the SVM model in the gender-dependent scenario using MFCC features. Figures 20 and 21 illustrate the confusion matrices obtained with the CNN model in the gender-dependent scenario using MFCC and Mel-spectrogram features, respectively. These figures also show that the performances of male and female genders are different, and lower bias is seen by using Mel-spectrogram features in the CNN model.

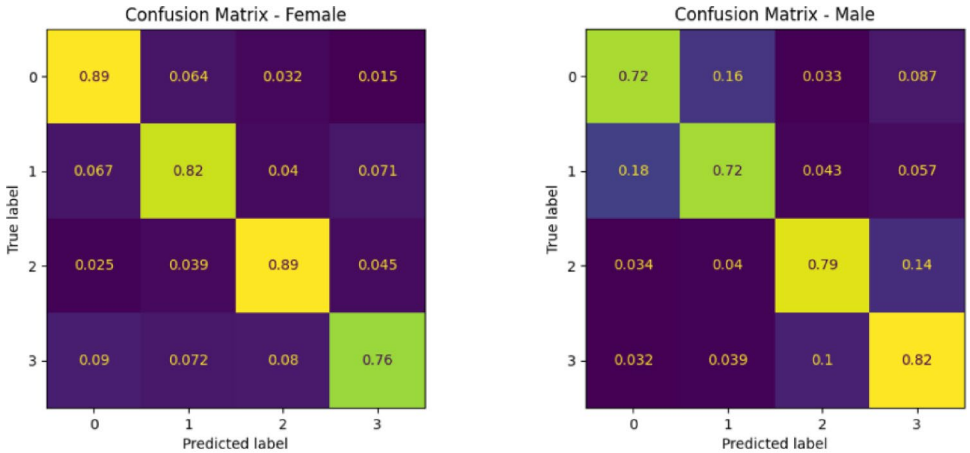
Table 16 Comparison of model performance between Shemo [16] and AutESD datasets in terms of UAR percentage

	Female		Male		Mean UAR	STD UAR
	Mean	STD	Mean	STD		
SVM (Shemo)	59.4		57.6		58.2	-
KNN (Shemo)	47.4		45.6		46.6	-
SVM (AutESD)	49.4	2.5	57.30	1.20	52.01	1.7
KNN (AutESD)	43.0	5.1	47.6	2.5	45.8	3.5
MFCC + CNN (AutESD)	75.2	2.0	71.0	4.1	72.6	3.00
MelSpec + CNN (AutESD)	78.5	4.0	75.7	4.0	76.8	4.00
MFCC + Transformer (AutESD)	76.7	6.0	73.8	3.7	74.9	4.6
MelSpec + Transformer (AutESD)	78.0	1.3	74.0	3.9	75.1	4.4



(a) (b)

Fig. 19 Confusion matrices for the SVM model using MFCC features in the gender-dependent scenario. **a** Female, **b** Male and class labels are encoded as: H = 0, A = 1, S = 2, and N = 3



(a) (b)

Fig. 20 Confusion matrices for the CNN model using MFCC features in the gender-dependent scenario. **a** Female, **b** Male and class labels are encoded as: H = 0, A = 1, S = 2, and N = 3

Figure 22 illustrates the learning curves for CNN-based and transformer models using both MFCC and Mel spectrogram features. It shows that we have set the hyperparameters well and no overfitting has occurred.

5.3.6 Discussion

According to Table 16, after evaluating four models of SVM, KNN, DNN (CNN-based), and Transformer on AutESD, we could achieve the highest UAR (unweighted-average-recall) with the DNN (CNN-based) + Mel Spectrogram model. This is due to the appropriate feature extraction acquired from CNN layers in the DNN model structure as

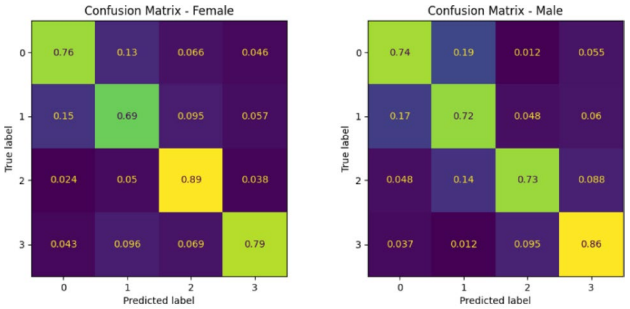


Fig. 21 Confusion matrices for the CNN model using Mel-spectrogram features in the gender-dependent scenario. **a** Female, **b** Male and class labels are encoded as: H = 0, A = 1, S = 2, and N = 3

shown in Fig. 16. Please note that these results occurred in gender-dependent case considering female and male classes separately in Table 16. However, in the gender-independent case without different training procedures for different genders, CNN achieved an UAR of 79% for MFCCs which is higher than Mel spectrogram features with 76.6% according to Table 14. The higher UARs achieved for females in Table 16 shows that they urged themselves imitating the emotions better even though the number of female speakers was fewer than male ones. According to Table 16, the classic models of SVM and KNN performed better for males, while DNN and transformer methods as examples of deep networks showed higher performance for females. This depicts that gender plays an important role in SER systems performance. Owing to the fact that there are acoustic differences between male and females, gender-dependent emotion recognition is more effective than gender-independent scenario. In fact, male and female voices are physiologically different because of differences in pitch and vocal tract length. Therefore, the trend of various models for male and female speakers could not be predicted, and they may have different performances, as seen in this paper. On the other hand, the learning structure is completely different for DNNs and SVM. CNNs could extract complex patterns from data, while SVM is an optimal margin classifier trying to find a margin between classes. Hence, we believe that CNNs may be capable of finding fine differences between genders better although this could not be generalized in all cases. These facts have led to different trends of models for male and female speakers in this work. In addition, according to Table 16, not much difference could be seen between male and female UARs representing low bias in terms of genders. The higher UARs and accuracies achieved by DNN (CNN-based) model compared to classic models of SVM and KNN is due to the fact that CNN layers are capable of extracting appropriate features for the classification purpose much better than classic models without further feature extraction and processing steps.

Additionally, according to the confusion matrices in Figs. 14 and 15, and 17, it is evident that obtaining high accuracy in the sad label in all three models has assigned almost the highest value. This is due to the nature of sadness emotion for imitation. In fact, based on the feedback

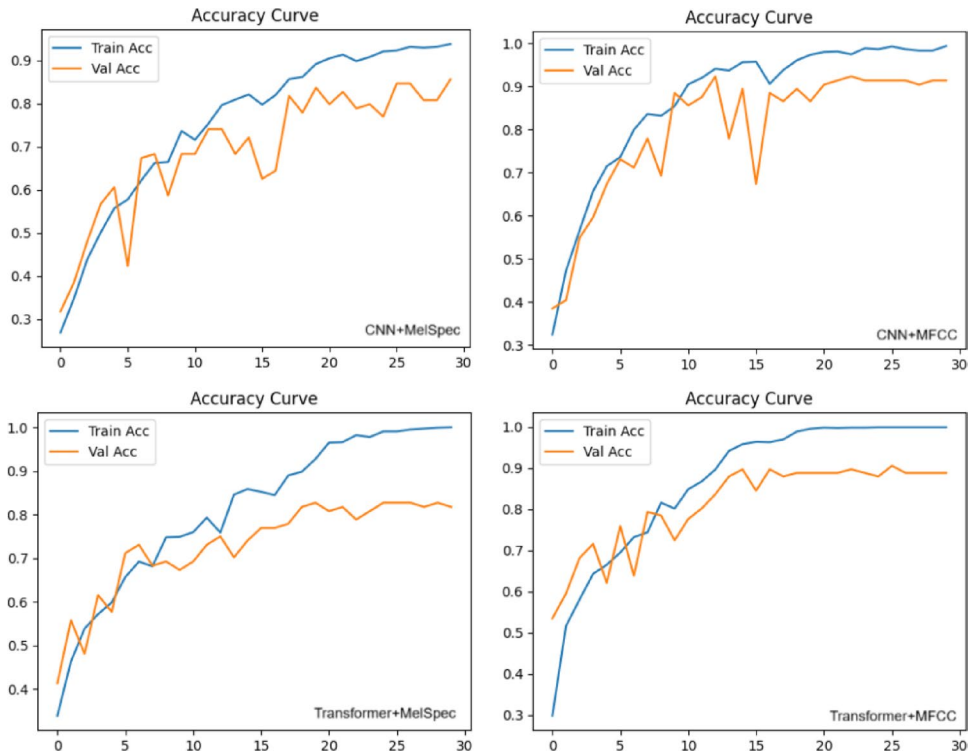


Fig. 22 Training and Validation curves for both CNN and transformer models. Top: CNN model with Mel spectrogram and MFCC, Bottom: transformer model with Mel spectrogram and MFCC

we got from the speakers, they were more confident imitating sadness imagining that they are depressed. Sadness is naturally very close to neutral case which could be mimicked very easily because it does not need any special emotion. Therefore, sadness could be mimicked well due to two reasons of easily simulating depression emotion as well as its closeness to neutral case. In addition, neutrality is mostly confused by sadness as expected. We believe that sadness class achieved the highest value in all models considering the fact that the speakers tried their best imitating an emotion instead of neutrality itself. Hence, they imitated sadness better than neutrality. The next highest correct classification rate is related to happy or angry classes, which could be imitated more easily than other classes due to their strong-emotion nature, and their easy imitation for non-professionals. The most misclassification occurred in neutral and angry classes. The reason for high misclassification in neutrality is the same as our justification for high classification rate in sadness. Actually, since the non-professionals goal was imitating the emotions well, they could achieve this goal in the sadness class the most, and due to its natural similarity to neutral class, it is mostly misclassified. As expected, the neutral class is mostly confused by sadness, while the most confusion for anger happens for happy class. Happy and angry emotions are naturally confused since they have high arousal values indicating strong emotions which could be easily confused even in acted datasets. It is worth mentioning that the high accuracy of the sadness class in all models also corresponds to the speaker-independent case we have performed in all models for the semi-natural gathered dataset from non-professionals. In other words, the speaker-independent case focuses on the nature of emotions among all speakers instead of the characteristics of each speaker pronouncing words individually.

It is important to note that the high quality and reliability of the labels in the AutESD dataset are supported by the developed labeling software. The software uses a two-level annotation scheme (Strong and Weak) with a "None" option for ambiguous or unclear cases, which allows annotators to express subtle or secondary emotions. Built-in mechanisms prevent duplicate labeling and automatically save annotations, ensuring each sample is uniquely labeled. Moreover, the system supports multi-user and remote participation, enabling a larger number of annotators and improving the consistency and reliability of the dataset. This careful labeling procedure likely contributes to the observed high accuracy in certain classes, such as neutral, across all models. Additionally, the combination of Strong and Weak labels provides insights for optimizing annotation strategies in semi-natural emotional speech datasets.

To further assess the quality of the emotion annotations in the AutESD dataset, we evaluated inter-rater agreement using Fleiss' Kappa across all annotators as shown in Table 6. For Strong emotions, Fleiss' Kappa ranged from 0.78 to 1.00 (average $\approx$ 0.84), indicating substantial to almost perfect agreement. For Weak emotions, Kappa ranged from 0.74 to 0.88 (average $\approx$ 0.78), showing strong agreement. When analyzing overlapping groups of 4–10 voters, Kappa remained between 0.70 and 0.90, confirming consistent labeling behavior across participants. These results validate the reliability of the dataset and suggest that the high quality of the annotations contributed to the strong performance observed in model evaluations.

According to the training and validation curves shown in Fig. 22, we have set the hyperparameters and model settings well and no overfitting has occurred for neither DNN nor transformer models.

Finally, it is necessary to mention that our sole objective in this paper has been focused on dataset collection for Persian speech emotion detection. Thus, we have designed several models solely for evaluating the dataset's performance showing its appropriateness for the emotion detection goal. Thus, we have chosen both classic and DNN models to show this. Furthermore, it is obvious that a larger dataset could lead to better results though creating such a dataset is more challenging and time-consuming in both steps of data collection and labeling.

6 Conclusion

Natural speech from non-professional speakers is crucial for practical speech processing applications. To achieve this, in this paper, we created a semi-natural emotional dataset called AutESD for SER task by recording emotional speech of 28 native students as non-professionals. AutESD contains 1428 Persian speech utterances in four emotions of neutrality, happiness, anger, and sadness recorded from speakers in different sessions to minimize fatigue effects. The use of non-professional speakers ensured accent and gender diversity while minimizing data collection costs. All of them speak Persian as their native language, but they might have minor accents to preserve the variety of ways that the Persian language is expressed in various locales. Thus, this gathered dataset could be considered as an acceptable resemblance to the real-world conditions such as on-line class evaluations. Since the data is semi-natural, we should have a labeling procedure to omit erroneous labels caused by non-professional speakers who could not imitate the emotion well or other problems occurred during recordings. Therefore, to facilitate labeling and create a structure for reducing labeling costs, and time, as well as increasing its accuracy, we proposed and developed a labeling software. To this end, a custom labeling tool was developed for annotating strong

and weak emotions, involving 16 annotators including both speech-experts and non-experts. This software also provided the annotators to do the labelling procedure remotely. It is worth stating that the participants in this labeling procedure were unaware of the true emotion label. Inter-rater reliability, measured by Fleiss' Kappa, ranged from 0.784 to 1.000 for strong and 0.739–0.876 for weak emotions, indicating substantial to almost perfect agreement. Baseline evaluations on speaker-independent cases considering both gender-independent and gender-dependent (training female and male genders separately) scenarios demonstrated strong performances with the highest UARs achieved for DNN-based models using either MFCCs or Mel spectrogram features. These results confirmed the dataset's validity and applicability for Persian SER research. Future work could include extending the dataset and integrating multimodal cues (speech and facial emotions) to further enhance emotion recognition robustness. Moreover, exploring more emotions (such as fear, surprise, stress, etc.) could also be investigated in future.

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Author contributions AR: data curation, conceptualization, methodology, software, writing; SS: data curation supervisor, conceptualization, methodology, writing, revision; ZSJ: data curation, writing.

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Data availability The dataset generated and analyzed during the current study is available on our institute's website, Speech and Signal Processing Research Lab, Department of Electrical Engineering, Amirkabir University of Technology, Tehran, Iran: [http://ele.aut.ac.ir/~sprl/AutEsd/AutESD.rar]. The dataset comprises speech recordings labeled for Persian speech emotion recognition, including four emotional states: neutrality, anger, happiness, and sadness. The recordings were collected from both male and female of 28 non-professional speakers who voluntarily participated in multiple sessions to minimize fatigue effects. Each participant read specifically designed sentences in the intended emotional tone to construct a semi-natural speech emotion recognition (SER) corpus. All data collection sessions were conducted in Amirkabir University of Technology acoustic room to ensure sound quality and minimize background noise. Participants were informed of their right to withdraw at any time. This dataset called AutESD could also be adapted to be used for on-line academic evaluations in future since we have gathered it from both male and female students in the age range of 23 to 30. All of the participants were native Persian speakers with a complete Persian accent. However, people were chosen from different cities in Iran in order to preserve the variety of ways that the Persian language is expressed in various locales. 28 speakers participated in the database recording (17 males and 11 females).The Sony F-VK98 microphone was used to collect the data in the acoustic room of our institute. Also, according to Tables 1 and 2, a sheet containing the texts was given to the speakers (There is a separate text for male and female speakers) so that the speakers could utter the sentences in target emotions. Moreover, in order to convey emotions better, we trained the speakers before starting the recordings. To achieve this, similar audios were played to arouse emotions in the speakers and adapt them. It should be mentioned that three texts were considered for each speaker. Out of these three sentences, two sentences were mandatory and the speaker has the option of selecting the third sentence from two optional sentences to have a better imitated emotion by each speaker.

Code availability We created AutESD dataset in this paper as a part of the Master's thesis in [38] under the permission of Amirkabir University of Technology. All materials supporting the findings of this study are publicly available at the following repositories: Code repository: https://github.com/alireza0r/AutESD. Dataset repository: http://ele.aut.ac.ir/~sprl/AutEsd/. Labeling software: https://github.com/alireza0r/EmotionalSpeechLabeling.

Declarations

Conflict of interest The authors declare no competing interests.

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